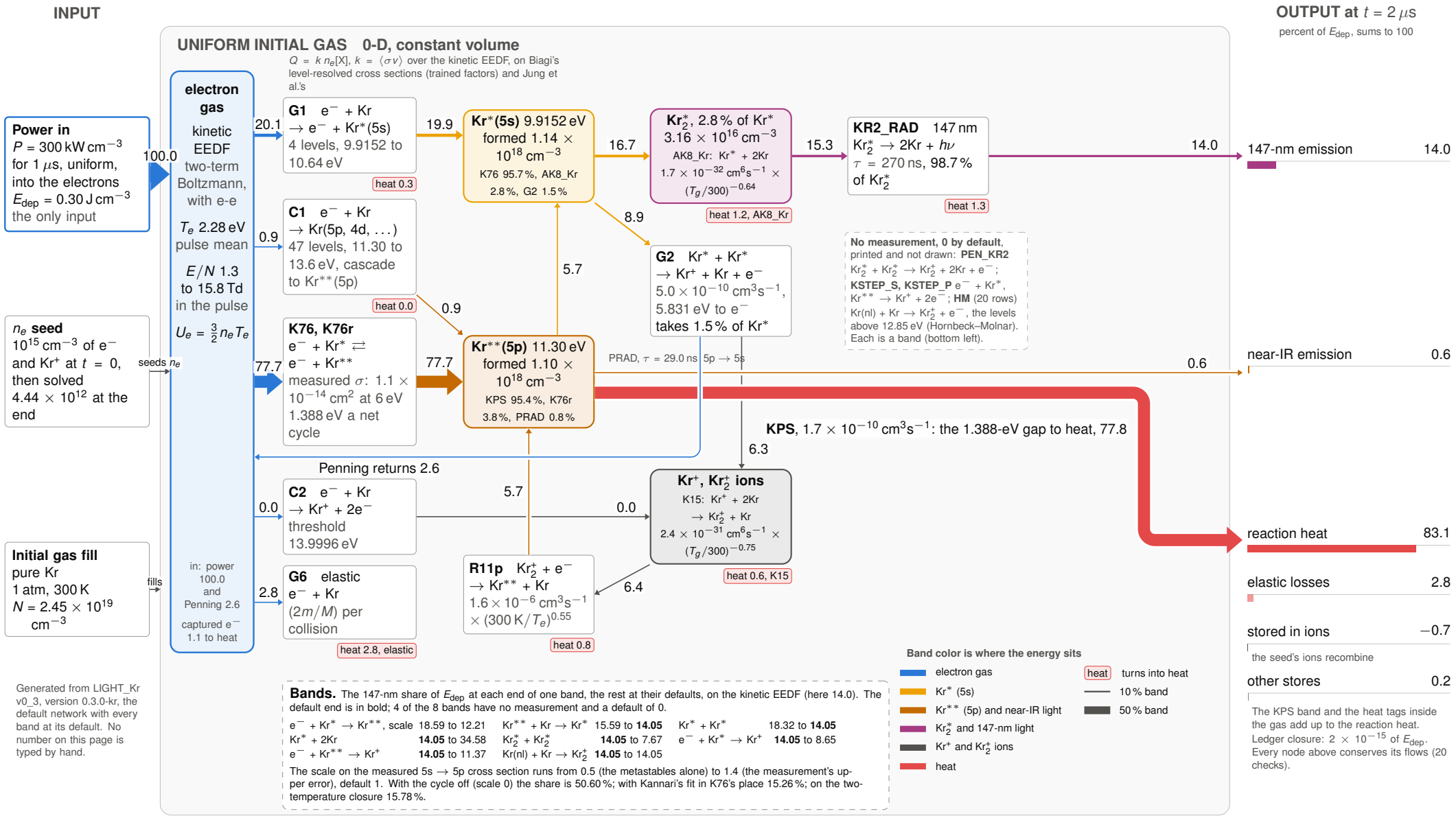
LIGHT_Kr system map, v0_3: energy flow through pure krypton

Reference point: 1 atm of Kr at 300 K, P = 300 kW cm^{-3} into the electrons for 1 μs so that E_{dep} = 0.30 J cm^{-3}, n_e seeded at 10^{15} cm^{-3} and then solved, every band at its default, run to 2 μs on the kinetic EEDF. Band width is proportional to the energy flow. Numbers are percent of E_{dep}, rounded to 0.1.



The outputs as flows, which overlap: to Kr* and Kr** by electron impact 21.0 %, to Kr+ 6.3 %, to Kr_2^* 14.2 % (formed times the 147-nm quantum), heat 83.1 %, elastic losses 2.8 %. **The 5s ↔ 5p cycle.** The electrons pump Kr* up to the 5p rung (K76, on Jung et al.'s measured cross sections) 34 times for each Kr_2^* formed, and krypton relaxes the rung back (KPS), so the cycle is drawn as its two net bands: of the Kr* formed (gross), 95.7 % is pumped up; the rung relaxes 95.4 % of it on Kr, radiates 0.8 % and the electrons return 3.8 %; the 1.388 eV of every net cycle, 1.05 × 10^{18} cm^{-3} of them, takes 77.7 % of the input from the electrons, and KPS turns it into heat. Penning ionization (G2) makes 100 % of the ions, and they recombine onto the rung (R11p). **The kinetic EEDF:** 381 solutions, 276 in the pulse; over its second half the field carries 1.0001 to 1.0002 of the input and the kinetic and the model's energy balances agree to 9 × 10^{-6} of it. At the end T_g = 808 K, p = 2.69 atm, n_e = 4.44 × 10^{12} cm^{-3}. **Sources** of the rates printed: cross sections Biagi, Magboltz 8.97, Nucl. Instrum. Methods A 421, 234 (1999), doi:10.1016/S0168-9002(98)01233-9, via LXCat, Pancheshnyi et al., Chem. Phys. 398, 148 (2012), doi:10.1016/j.chemphys.2011.04.020, times factors trained on krypton's measured drift velocities and ionization coefficients (the 5s excitation ×0.89, the 5p ×0.92, the higher levels ×0.75, the momentum transfer ×0.93 to ×1.23); K76 Jung et al., Phys. Rev. A 73, 022722 (2006), doi:10.1103/PhysRevA.73.022722; AK8_Kr Ueda et al., Rev. Laser Eng. 12, 357 (1984), doi:10.2184/lsj.12.357, on Wieme and Lenaerts' law, J. Chem. Phys. 74, 483 (1981), doi:10.1063/1.440855; KR2_RAD Morikawa et al., J. Chem. Phys. 91, 1469 (1989), doi:10.1063/1.457108; G2 Leichner and Ericson, Phys. Rev. A 9, 251 (1974), doi:10.1103/PhysRevA.9.251, as Kannari et al. print it, J. Appl. Phys. 57, 4309 (1985), doi:10.1063/1.334590; KPS Gazeli et al., Phys. Plasmas 28, 043301 (2021), doi:10.1063/5.0041471; PRAD Dzierżęga et al., Phys. Rev. A 62, 022505 (2000), doi:10.1103/PhysRevA.62.022505; R11p Shiu and Biondi, Phys. Rev. A 16, 1817 (1977), doi:10.1103/PhysRevA.16.1817; K15 the mean of Bhattacharya, Phys. Rev. A 1, 1196 (1970), doi:10.1103/PhysRevA.1.1196, Grössl et al., J. Chem. Phys. 74, 1728 (1981), doi:10.1063/1.441261, and Smith et al., J. Phys. B 5, 2134 (1972), doi:10.1088/0022-3700/5/11/020.